MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

MIL-STD-1325-171 (NAVY)

BOMB, GENERAL PURPOSE, M117 (750 LB)

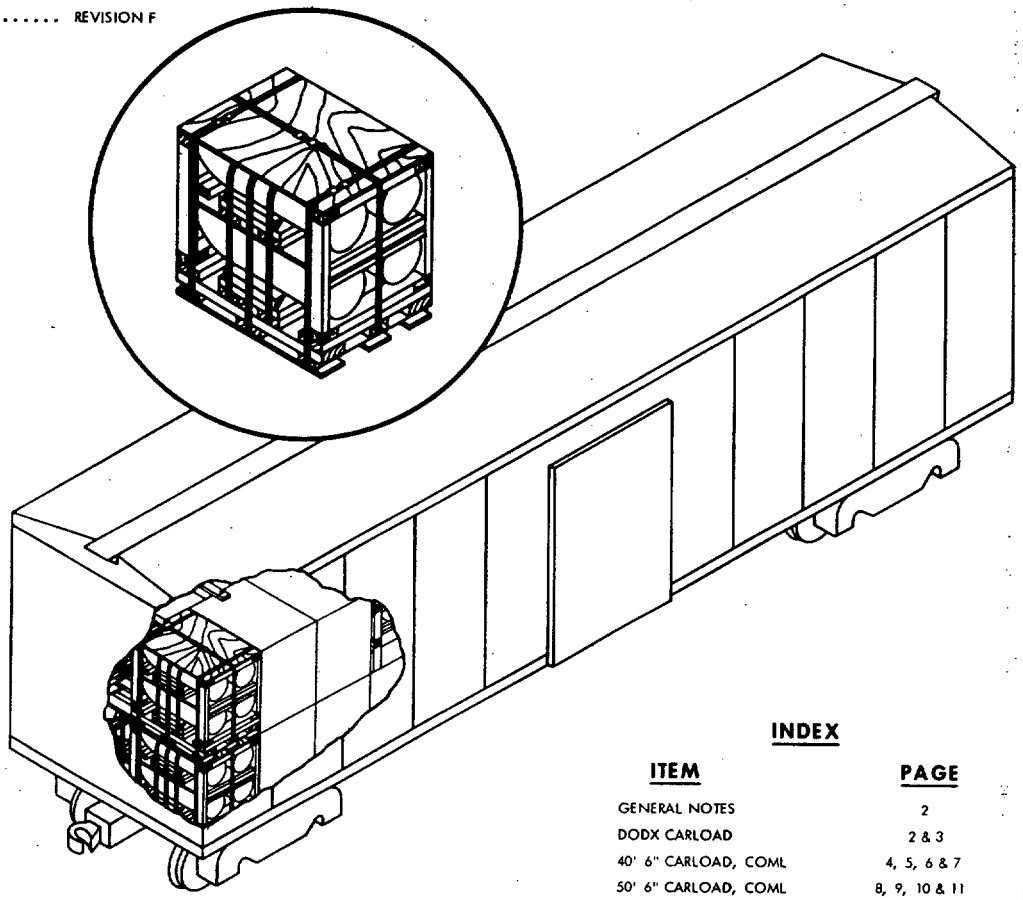
27 AUGUST 1974

AIR FORCE UNIT LOAD

TPO 1325-232-6331 REVISION F

UNIT LOAD DATA

WEIGHT (APPROXIMATE) ..... 3124 LBS.
DIMENSIONS ..... 49 1/4 L x 32 1/4 W x 42 3/4 H
CUBE ..... 39.3 CU.FT.
DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A
AIR FORCE DRAWING NO.
1325-232-6331 ..... REVISION F



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DODX CARLOAD	2 & 3
40' 6" CARLOAD, COML	4, 5, 6 & 7
50' 6" CARLOAD, COML	8, 9, 10 & 11
50' 6" CARLOAD, COML (USING LOAD DIVIDERS)	12, 13, 14 & 15
DUMMY UNIT LOAD	16

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M S Gray NWHL 8/2/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J E Kelly 8/2/74
SIGNATURE AIRSYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

SUPERVISOR, MILITARY & INTERMODAL SERVICES

A. F. Grassmuck 6/24/74
SIGNATURE DATE

ORIGINATOR Charles McBride 8/1/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY

NAVAL AMMUNITION DEPOT, EARLE, NEW JERSEY

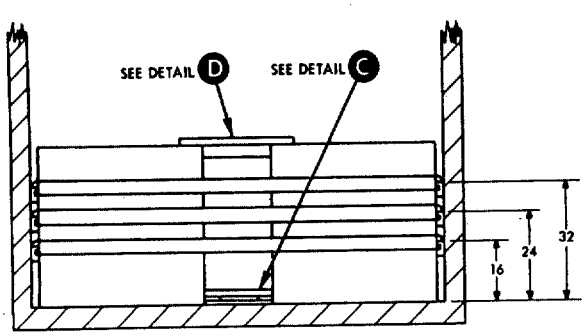
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GENERAL NOTES

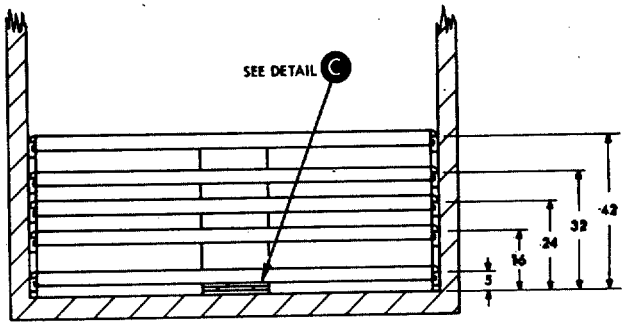
1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 750 LB GENERAL PURPOSE BOMBS M117 PALLETIZED IN ACCORDANCE WITH AIR FORCE DRAWING NO. 1325-232-6331 REVISION F.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. 1).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2500 LBS OF THE LADING.



SECTION A-A
SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS



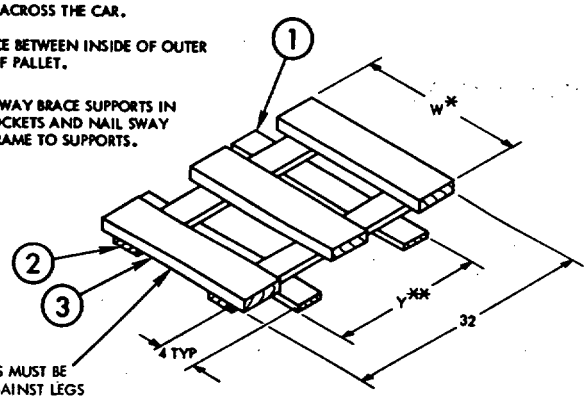
SECTION B-B
SHOWING LOCATION OF DOORWAY MEMBERS
USED FOR CROSS MEMBERS

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*W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

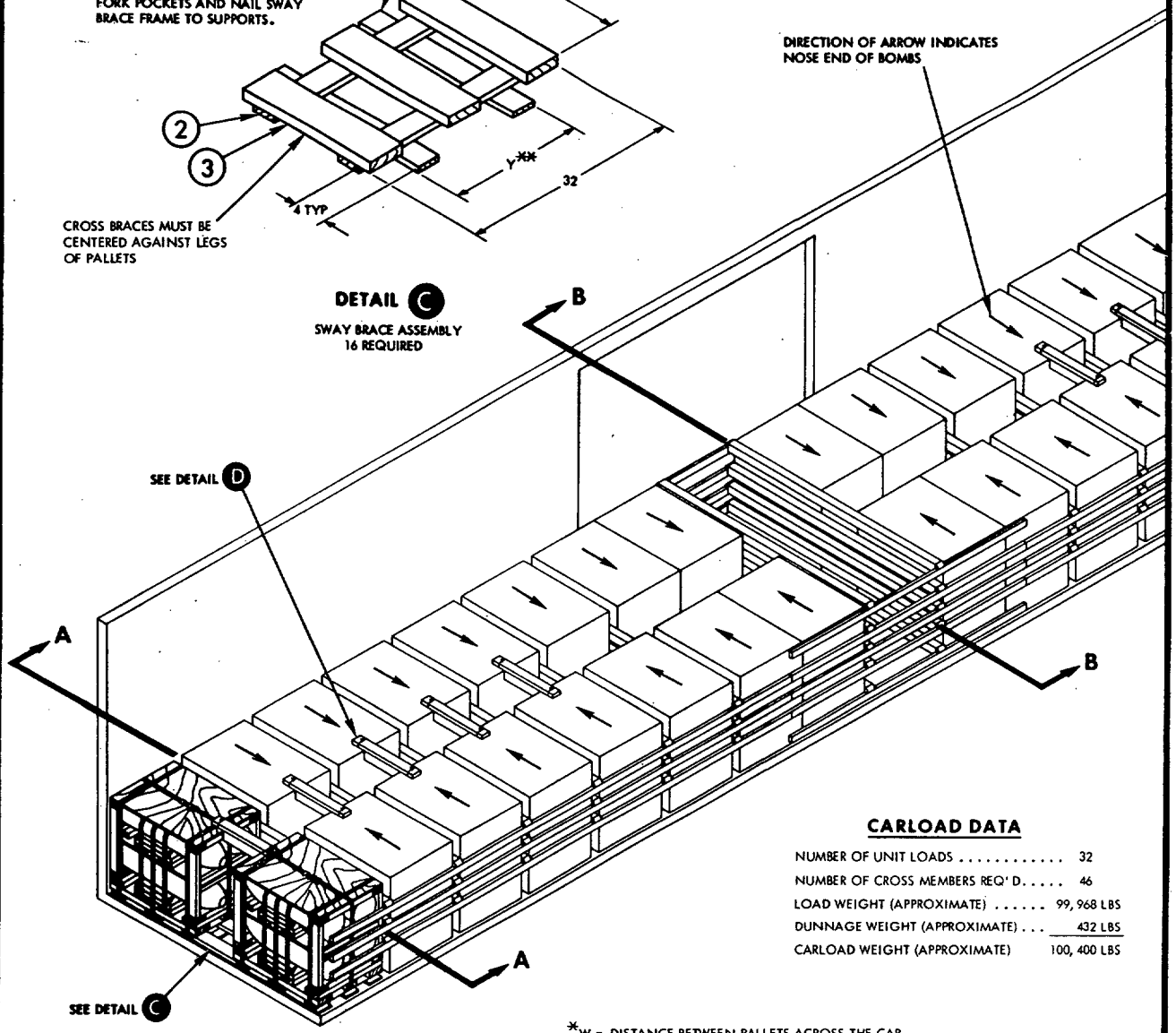
**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS

DETAIL C
SWAY BRACE ASSEMBLY
16 REQUIRED



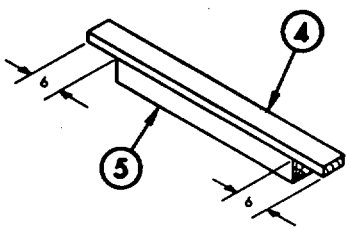
CARLOAD DATA

NUMBER OF UNIT LOADS	32
NUMBER OF CROSS MEMBERS REQ'D	46
LOAD WEIGHT (APPROXIMATE)	99, 968 LBS
DUNNAGE WEIGHT (APPROXIMATE)	432 LBS
CARLOAD WEIGHT (APPROXIMATE)	100, 400 LBS

*W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

**PIECE 2 NAILS TO PIECE 3 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 1 WITH 6d NAILS, 1 PER JOINT.

***SECURE THE TOP SWAY BRACE WITH NO. 14 GAGE TIE WIRE 24" LONG. ENCIRCLE THE SUPPORT PIECE AND THE STRAP OF THE UNIT LOAD AND TWIST TAUT.



DETAIL D
TOP SWAY BRACE
10 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
5	CROSS BRACE	4 x 4 x CUT TO FIT	10	SEE 4	-	-
4	SUPPORT PIECE	2 x 4 x CUT TO FIT	10	5	4	16d
				SEE NOTE ***		
3	CROSS BRACE	2 x 6 x W*	48	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 32	32	1, 3	SEE NOTE **	
1	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	32	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. TO COMPLY WITH GENERAL RULE 3 (E) IN AAR CIRCULAR NO. 42-D THE PARTIAL SECOND LAYER MAY ONLY BE LOADED IN BOXCARS BUILT ON OR AFTER 1 JANUARY 1966. IF THE PARTIAL SECOND LAYER IS NOT REQUIRED OR THE BOXCAR IN USE WAS BUILT PRIOR TO 1 JANUARY 1966 THE CARLOAD MAY CONSIST OF 28 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THE FIRST LAYER DUNNAGING PROCEDURES OF THIS DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.

* W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

** PIECE 2 NAILS TO PIECE 3 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 1 WITH 6d NAILS, 1 PER JOINT.

*** SECURE THE TOP SWAY BRACE WITH NO. 14 GAGE TIE WIRE 24" LONG, ENCIRCLE THE SUPPORT PIECE AND STRAP OF THE UNIT LOAD AND TWIST TAUT.

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

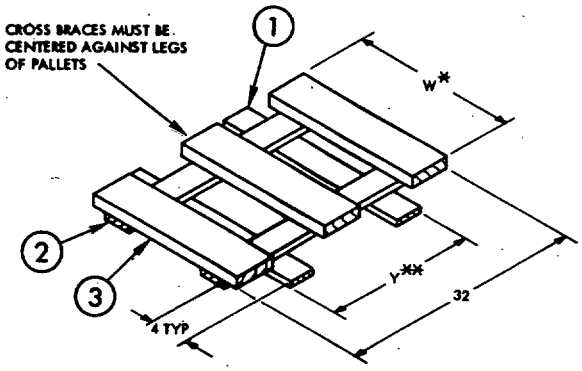
LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
28	FLOOR CLEAT	2 x 4 x (W+4)*	6	27	5	16d
27	FLOOR CLEAT	2 x 4 x W*	6	CAR FLOOR	5	16d
26	STAPLE	FOR 1 1/4 STRAP	4	4	2 STAPLES PER STRAP	
25	SEAL	1 1/4	4	-	-	-
24	DOORWAY PROTECTION STRAP	1 1/4 x .035 x 28 FT	2	-	-	-
23	GATE HOLDDOWN	2 x 4 x 28	2	21, 22	3 PER JOINT	10d
22	SPACER	1 x 6 x 18	2	21	4 PER JOINT	6d
21	VERTICAL SUPPORT	2 x 4 x 24	4	17	3 PER JOINT	10d
20	STRUT	4 x 4 x WEDGE FIT****	16	18	2 PER JOINT	16d
19	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	4	18	3 PER JOINT	10d
18	CENTER GATE VERTICAL	2 x 6 x 44	16	SEE 17	-	-
17	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	4	18	3 PER JOINT	10d
16	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	4	10d
15	INTERMEDIATE WALL CLEAT	2 x 4 x 14 1/2	4	CAR WALL	4	10d

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
14	HORIZONTAL BACKUP CLEAT	2 x 6 x 30	8	10	14	16d
13	DIAGONAL BRACE	4 x 4 x 50 SEE DETAIL G	8	8, 10	1 EACH END	60d
12	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	11	7	16d
11	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	10	7	16d
10	HORIZONTAL WALL CLEAT	2 x 6 x EXTEND ACROSS DOOR	4	CAR WALL	1 PER FOOT	10d
9	CENTER CLEAT	2 x 4 x 36	4	8	7	16d
8	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 7	-	-
7	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	8	2 PER FOOT	10d
6	LOWER WALL CLEAT	2 x 4 x 12	4	CAR WALL	4	10d
5	CROSS BRACE	4 x 4 x CUT TO FIT	12	SEE 4	-	-
4	SUPPORT PIECE	2 x 4 x CUT TO FIT	12	5	4	16d
				SEE NOTE***		
3	CROSS BRACE	2 x 6 x W*	54	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 32	36	1, 3	SEE NOTE**	
1	SWAY BRACE SUPPORT	1 x 4 x (W - 8)*	36	SEE 2	-	-

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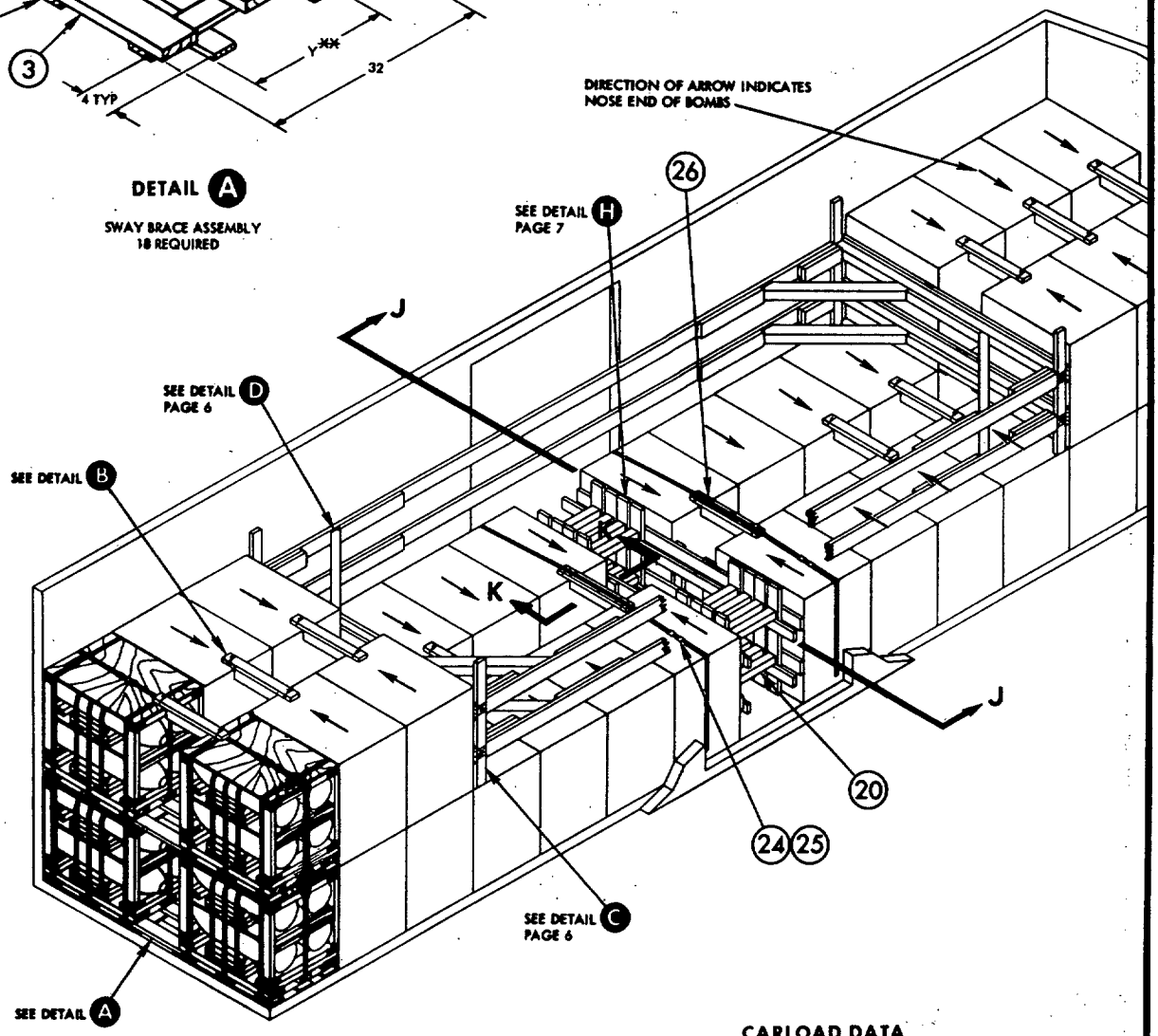
*W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

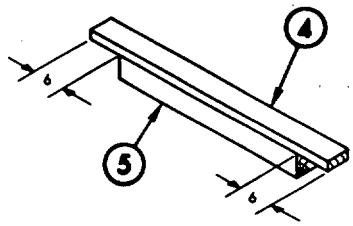
DETAIL A

SWAY BRACE ASSEMBLY
18 REQUIRED



CARLOAD DATA

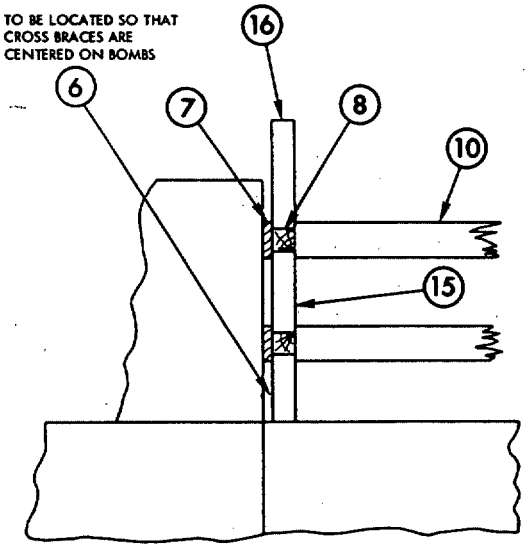
NUMBER OF UNIT LOADS	40
LOAD WEIGHT (APPROXIMATE)	124, 960 LBS
DUNNAGE WEIGHT (APPROXIMATE)	1, 823 LBS
CARLOAD WEIGHT (APPROXIMATE)	126, 783 LBS



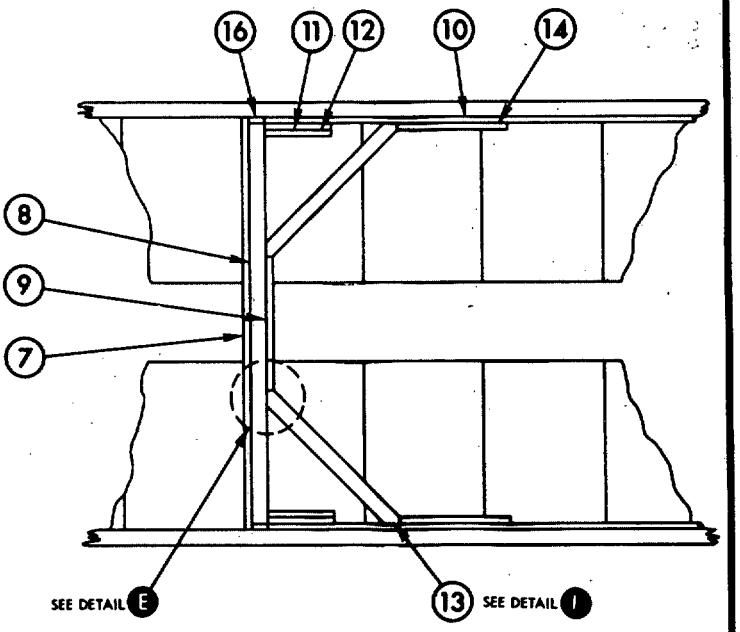
DETAIL B

TOP SWAY BRACE
12 REQUIRED

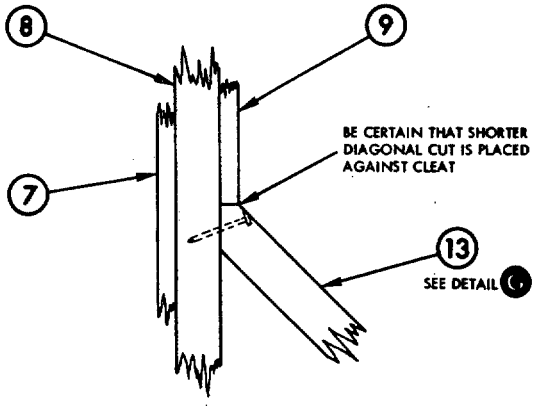
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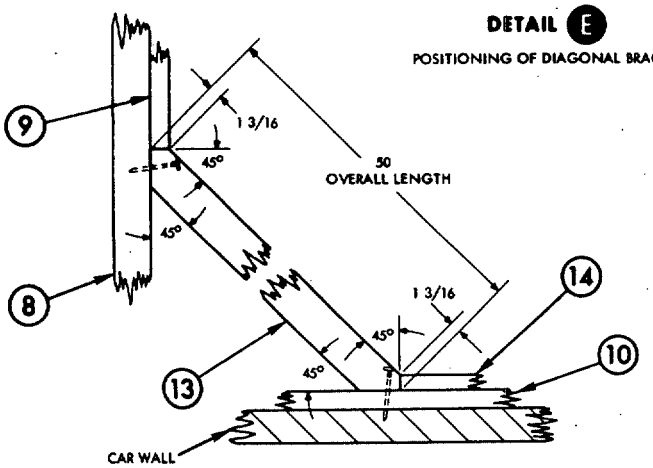
DETAIL C



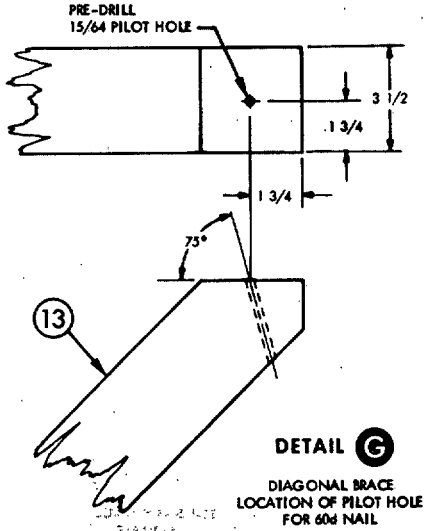
DETAIL D
PARTIAL LAYER BRACING



DETAIL E
POSITIONING OF DIAGONAL BRACE

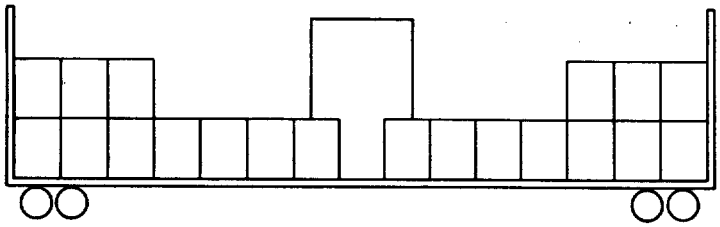


DETAIL F
POSITIONING OF DIAGONAL BRACE

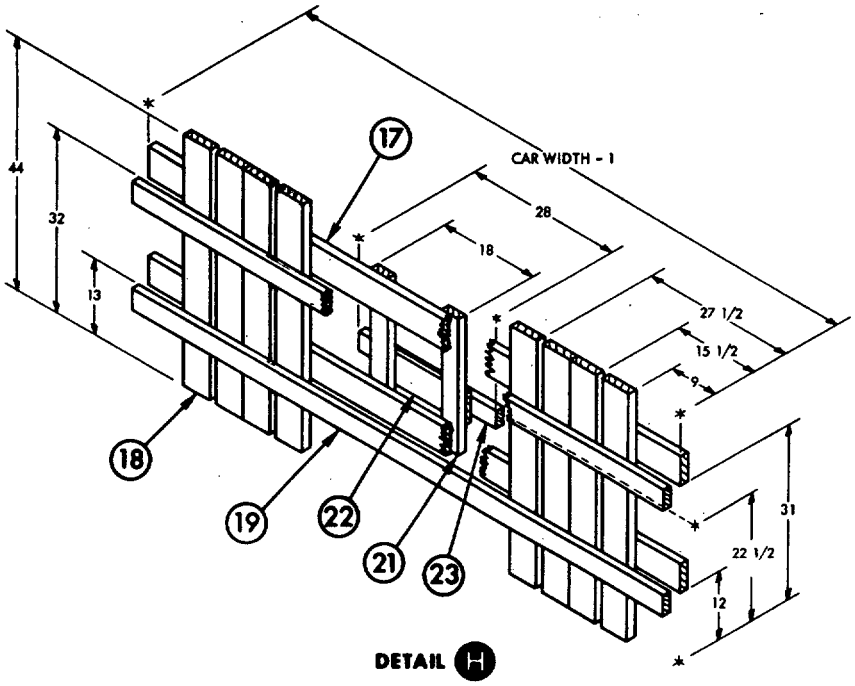


DETAIL G
DIAGONAL BRACE LOCATION OF PILOT HOLE FOR 60d NAIL

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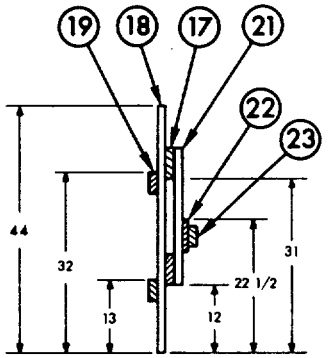


ELEVATION
LOADING PLAN

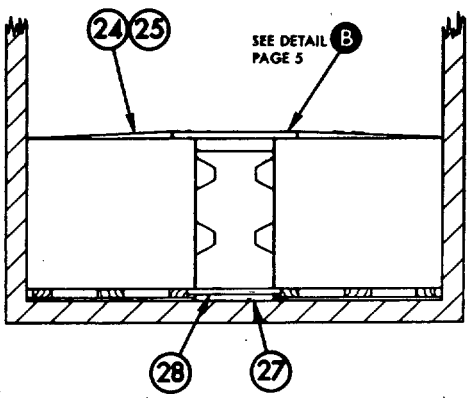


DETAIL H

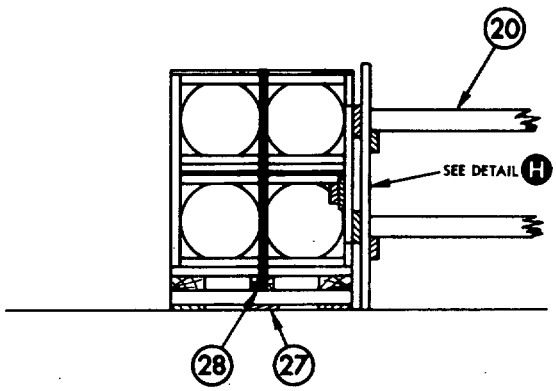
CENTER GATE
2 REQUIRED



END VIEW



SECTION J-J



SECTION K-K

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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF EITHER 46 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, OR 36 UNIT LOADS COMPRISING THE FIRST LAYER ONLY OF THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A SUPPORT PIECE AND CROSS BRACE, TWO PER STACK (SEE DETAIL B) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO SUPPORT PIECE WITH TWO 1 1/4" STAPLES. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.

* W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

** PIECE 2 NAILS TO PIECE 3 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 1 WITH 6d NAILS, 1 PER JOINT.

*** SECURE THE TOP SWAY BRACE WITH NO. 14 GAGE TIE WIRE 24" LONG. ENCIRCLE THE SUPPORT PIECE AND THE STRAP OF THE UNIT LOAD AND TWIST TAUT.

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

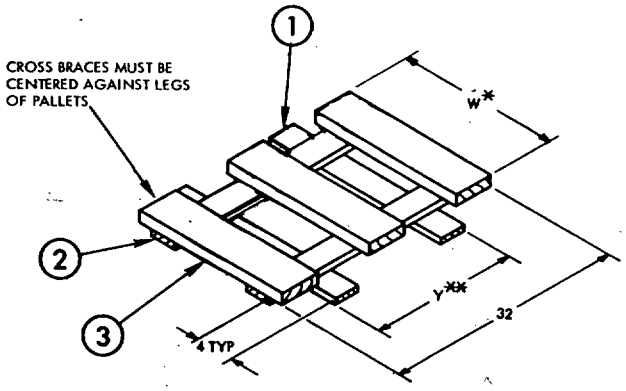
LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
28	FLOOR CLEAT	2 x 4 x (W + 4)*	6	27	5	16d
27	FLOOR CLEAT	2 x 4 x W*	6	CAR FLOOR	5	16d
26	STAPLE	FOR 1 1/4 STRAP	8	4	2 STAPLES PER STRAP	
25	SEAL	1 1/4	8	-	-	-
24	DOORWAY PROTECTION STRAP	1 1/4 x .035 x 28 FT	4	-	-	-
23	GATE HOLDDOWN	2 x 4 x 28	2	21, 22	3 PER JOINT	10d
22	SPACER	1 x 6 x 18	2	21	4 PER JOINT	6d
21	VERTICAL SUPPORT	2 x 4 x 24	4	17	3 PER JOINT	10d
20	STRUT	4 x 4 x WEDGE FIT ****	16	18	2 PER JOINT	16d
19	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	4	18	3 PER JOINT	10d
18	CENTER GATE VERTICAL	2 x 6 x 44	16	SEE 17	-	-
17	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	4	18	3 PER JOINT	10d
16	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	4	10d
15	INTERMEDIATE WALL CLEAT	2 x 4 x 14 1/2	4	CAR WALL	4	10d

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
14	HORIZONTAL BACKUP CLEAT	2 x 6 x 30	8	10	14	16d
13	DIAGONAL BRACE	4 x 4 x 50 SEE DETAIL J	8	8, 10	1 EA. END	60d
12	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	11	7	16d
11	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	10	7	16d
10	HORIZONTAL WALL CLEAT	2 x 6 x ENTEND ACROSS DOOR	4	CAR WALL	1 PER FOOT	10d
9	CENTER CLEAT	2 x 4 x 36	4	8	7	16d
8	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 7	-	-
7	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	8	2 PER FOOT	10d
6	LOWER WALL CLEAT	2 x 4 x 12	4	CAR WALL	4	10d
5	CROSS BRACE	4 x 4 x CUT TO FIT	14	SEE 4	-	-
4	SUPPORT PIECE	2 x 4 x CUT TO FIT	14	5 SEE NOTE ***	4	16d
3	CROSS BRACE	2 x 6 x W*	63	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 32	42	1, 3	SEE NOTE **	
1	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	42	SEE 2	-	-

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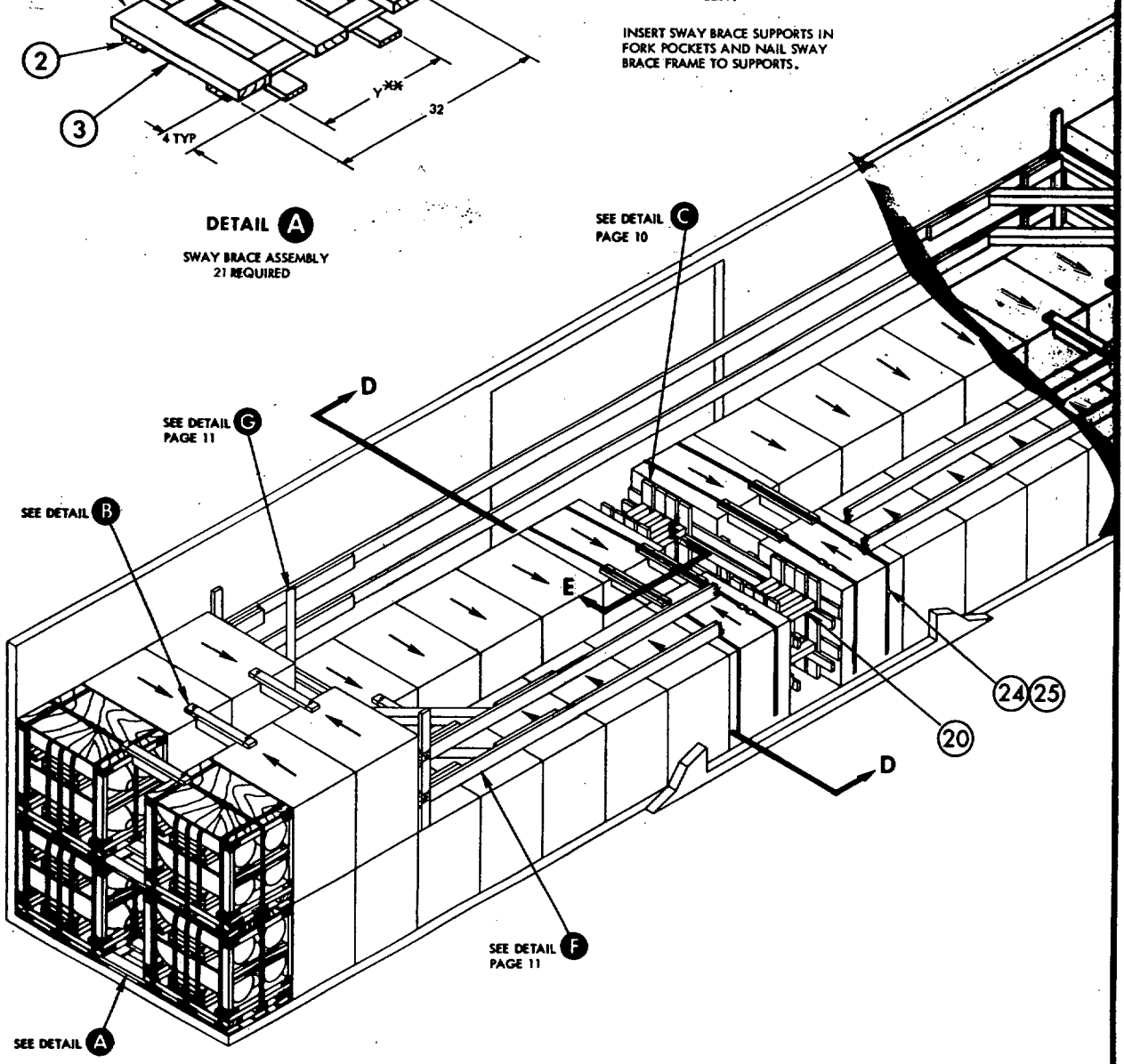
*W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

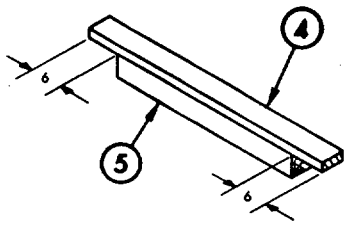
DETAIL A

SWAY BRACE ASSEMBLY
21 REQUIRED



CARLOAD DATA

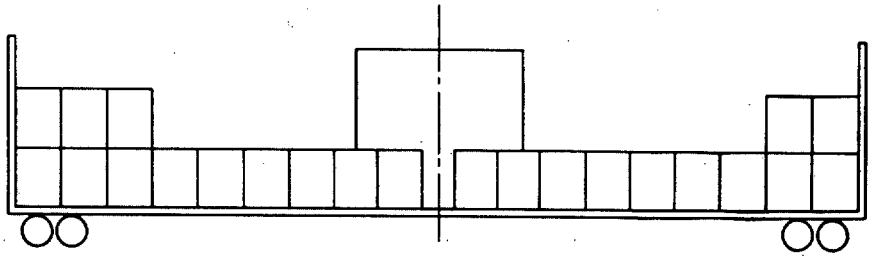
NUMBER OF UNIT LOADS	46
LOAD WEIGHT (APPROXIMATE)	143, 704 LBS
DUNNAGE WEIGHT (APPROXIMATE)	2, 138 LBS
CARLOAD WEIGHT (APPROXIMATE)	145, 878 LBS



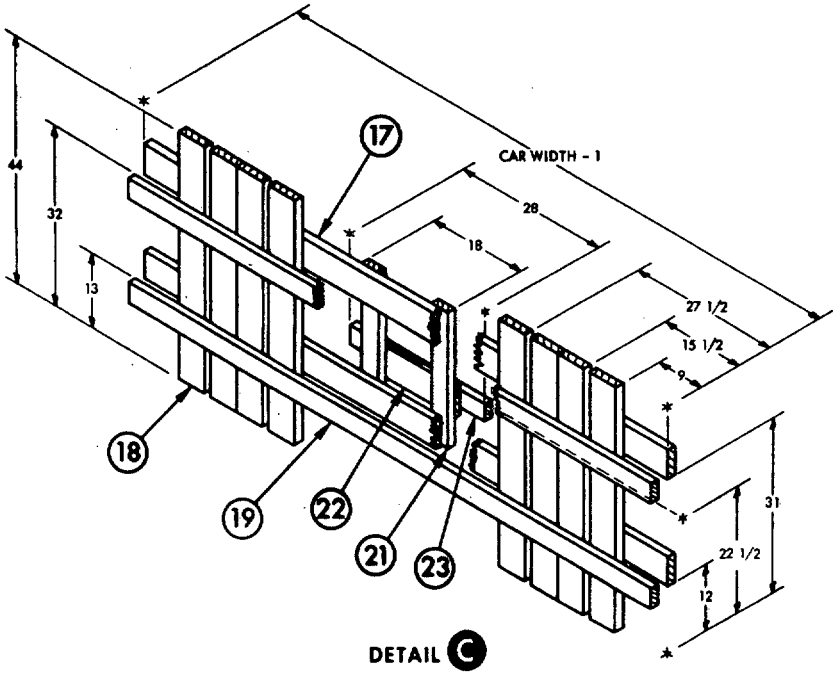
DETAIL B

TOP SWAY BRACE
12 REQUIRED

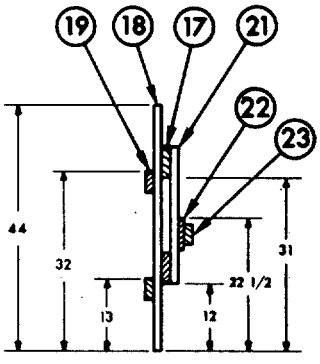
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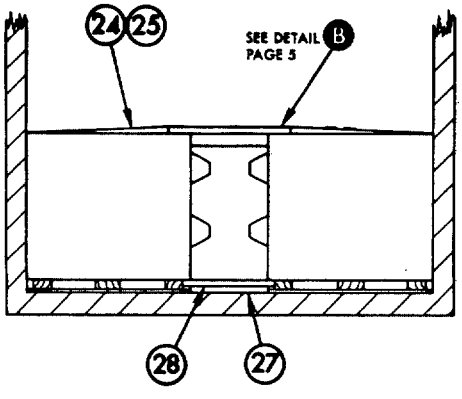
ELEVATION
LOADING PLAN



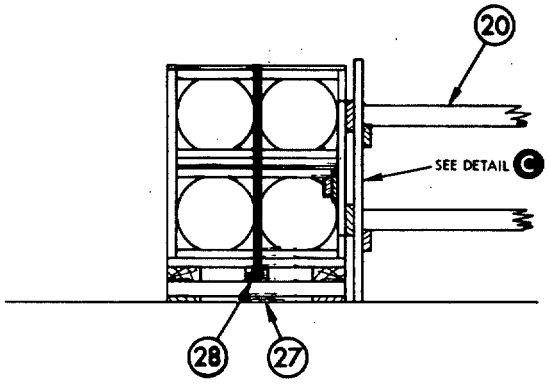
DETAIL C
CENTER GATE
2 REQUIRED



END VIEW

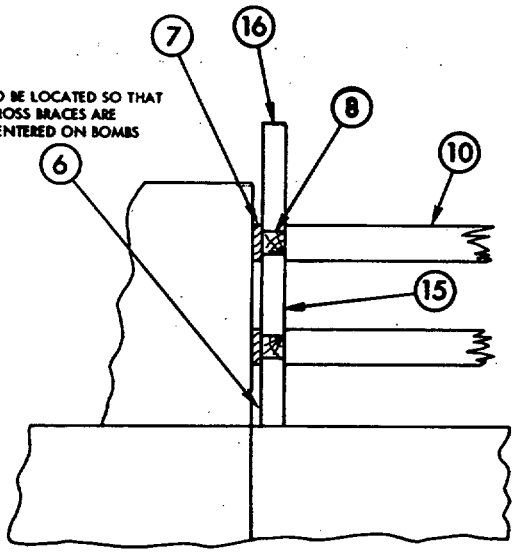


SECTION D-D

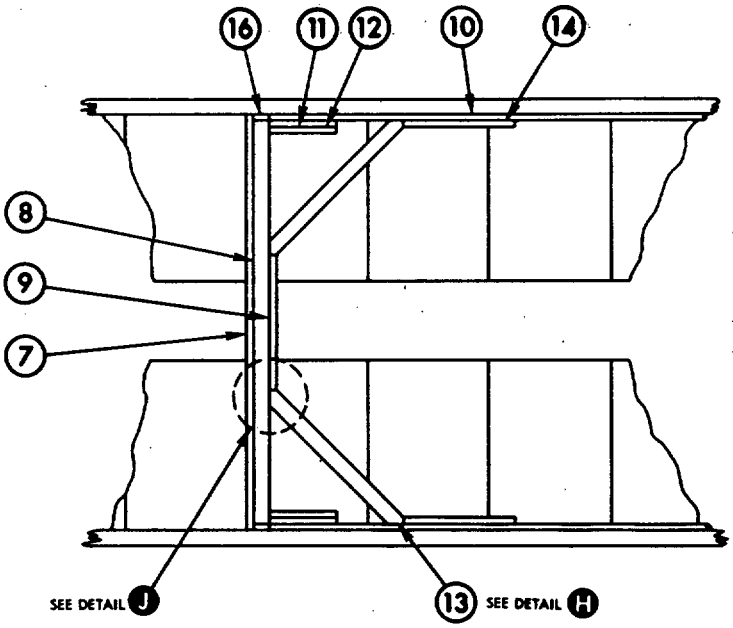


SECTION E-E

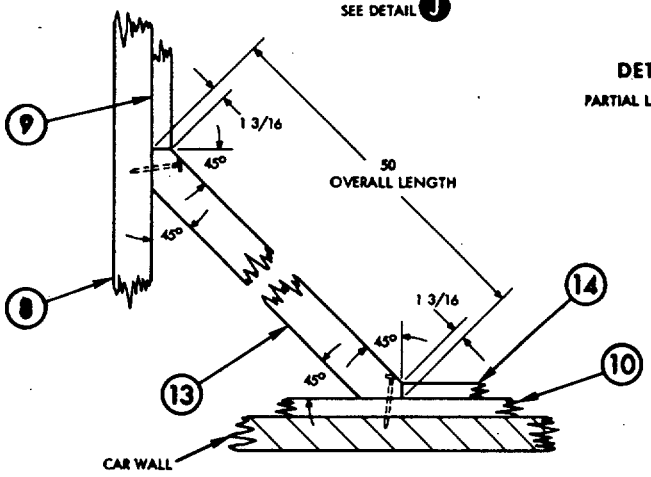
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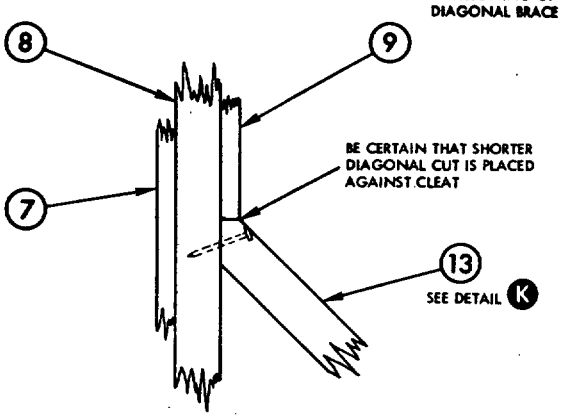
DETAIL F



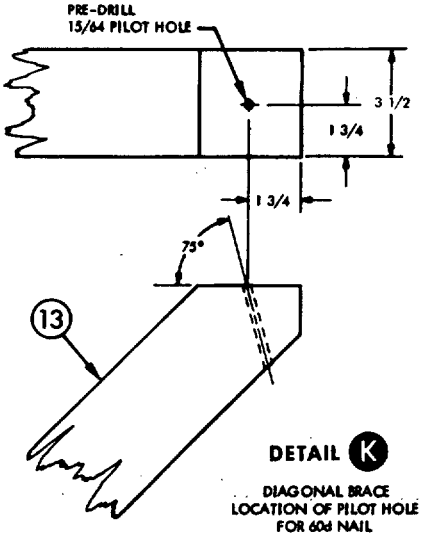
DETAIL G
PARTIAL LAYER BRACING



DETAIL H
POSITIONING OF DIAGONAL BRACE



DETAIL J
POSITIONING OF DIAGONAL BRACE



DETAIL K
DIAGONAL BRACE LOCATION OF PILOT HOLE FOR 60d NAIL

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50 FT 6 IN. BOXCAR, COMMERCIAL (USING LOAD DIVIDERS)

1. THESE PROCEDURES DEPICT THE METHOD OF LOADING THE AIR FORCE UNIT LOAD OF 750 LB. GENERAL PURPOSE BOMBS M117 PALLETIZED IN ACCORDANCE WITH AIR FORCE DRAWING NO. 1325-232-6331 REVISION F IN CUSHIONED BOXCARS WHICH ARE EQUIPPED WITH LOAD DIVIDERS AND WITH OR WITHOUT ADJUSTABLE SIDE FILLERS. ONLY CUSHIONED CARS THAT HAVE SLIDING CENTER SILL TYPE CUSHIONING DEVICES OR END-OF-CAR TYPE DEVICES WHICH HAVE AT LEAST 15 INCHES OF TRAVEL ARE ACCEPTABLE. (ONLY THOSE CUSHIONED BOXCARS EQUIPPED WITH LOAD DIVIDERS MANUFACTURED BY EVANS, EQUIPCO AND PRECO HAVE BEEN TESTED AND APPROVED BY THE AAR, BUX.)
2. BOXCARS EQUIPPED WITH ADJUSTABLE SIDE FILLERS THAT HAVE 3/8 INCH OR THICKER PANELS MAY BE USED. HOWEVER, THESE SIDE FILLERS MUST NOT BE USED FOR LATERAL BLOCKING. THEY MUST BE RETRACTED AND LOCKED AGAINST THE CAR SIDE WALL, AND A FILL PIECE INSTALLED IN THE VOID, IF ONE EXISTS BETWEEN THE CAR SIDE WALL AND THE SIDE FILLER PANEL. (SEE DETAIL G, PAGE 15.)
3. PRIOR TO LOADING A CAR EQUIPPED WITH LOAD DIVIDERS, A VERY CAREFUL INSPECTION MUST BE MADE TO ENSURE THAT THE CAR AND THE CAR EQUIPMENT ARE IN GOOD CONDITION. THE CONDITION OF THE LOAD DIVIDERS SHOULD BE CHECKED THOROUGHLY. BREAKS IN WELDING, BENT OR OTHERWISE DAMAGED LOCKING PINS, AND BENT OR DEFORMED BULKHEADS ARE REASONS FOR REJECTING CARS. AFTER THE LOAD DIVIDERS ARE POSITIONED AGAINST THE LADING, AND THE LOCKING PINS ARE ENGAGED IN THE HOLES OF THE RAILS, THE LOCKING PINS MUST BE INSPECTED TO ENSURE THAT THE PINS ARE FULLY ENGAGED IN THE LOCKING HOLES. IF THE LOWER PINS ARE NOT FULLY SEATED IN THE HOLES, THE LINKAGE MECHANISM SHOULD BE ADJUSTED AS REQUIRED SO THAT THE PINS WILL BE FULLY SEATED INTO THE LOCKING HOLES OF THE LOWER RAILS. IF PRESENT, DEBRIS MUST BE REMOVED FROM BENEATH THE LOCKING HOLES SELECTED FOR SECURING A DIVIDER.
4. THE CARLOAD CONSISTS OF 46 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
5. IF THE LOAD LIMIT OF THE CAR DOES NOT PERMIT LOADING 46 UNIT LOADS IN THE CAR, DUMMY UNIT LOADS, CONSTRUCTED AS SHOWN ON PAGE 16 MAY BE SUBSTITUTED TO LESSEN THE CARLOAD WEIGHT.
6. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
7. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 INCH STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.

* W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

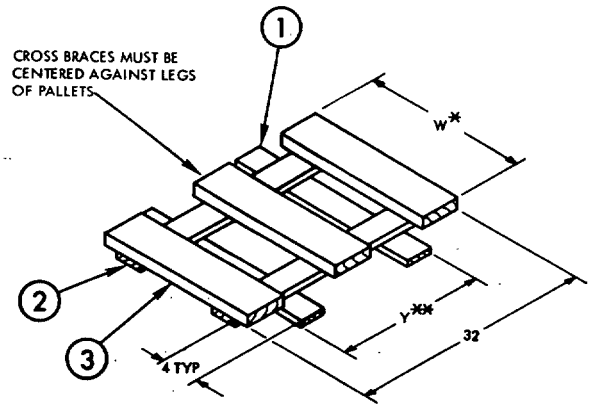
** PIECE 2 NAILS TO PIECE 3 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 1 WITH 6d NAILS, 1 PER JOINT.

*** SECURE THE TOP SWAY BRACE WITH NO. 14 GAGE TIE WIRE 24" LONG. ENCIRCLE THE SUPPORT PIECE AND THE STRAP OF THE UNIT LOAD AND TWIST TAUT.

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

					2 STAPLES PER STRAP		9	CENTER GATE VERTICAL	2 x 6 x 44	12	SEE 8	-	-
17	STAPLE	FOR 1 1/4 STRAP	8	4			8	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	4	9	3/JOINT	10d
16	SEAL	1 1/4	8	-	-	-	7	FLOOR CLEAT	2 x 4 x (W+4)*	9	6	5	16d
15	DOORWAY PROTECTION STRAP	1 1/4 x .035 x 28 FT	4	-	-	-	6	FLOOR CLEAT	2 x 4 x W*	9	CAR FLOOR	5	16d
14	GATE HOLDDOWN	2 x 4 x 28	2	12, 13	3/JOINT	10d	5	CROSS BRACE	4 x 4 x CUT TO FIT	14	SEE 4	-	-
13	SPACER	1 x 6 x 18	2	12	4/JOINT	6d	4	SUPPORT PIECE	2 x 4 x CUT TO FIT	14	5	4 SEE NOTES***	16d
12	VERTICAL SUPPORT	2 x 4 x 24	4	8	3/JOINT	10d	3	CROSS BRACE	2 x 6 x W*	60	SEE 2	-	-
11	STRUT	4 x 4 x WEDGE FIT****	12	9	2/JOINT	16d	2	SWAY BRACE LONGITUDINAL	1 x 4 x 32	40	1, 3	SEE NOTE**	
10	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	4	9	3/JOINT	10d	1	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	40	SEE 2	-	-
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE	PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS AND NAILING DATA							LIST OF MATERIALS AND NAILING DATA						

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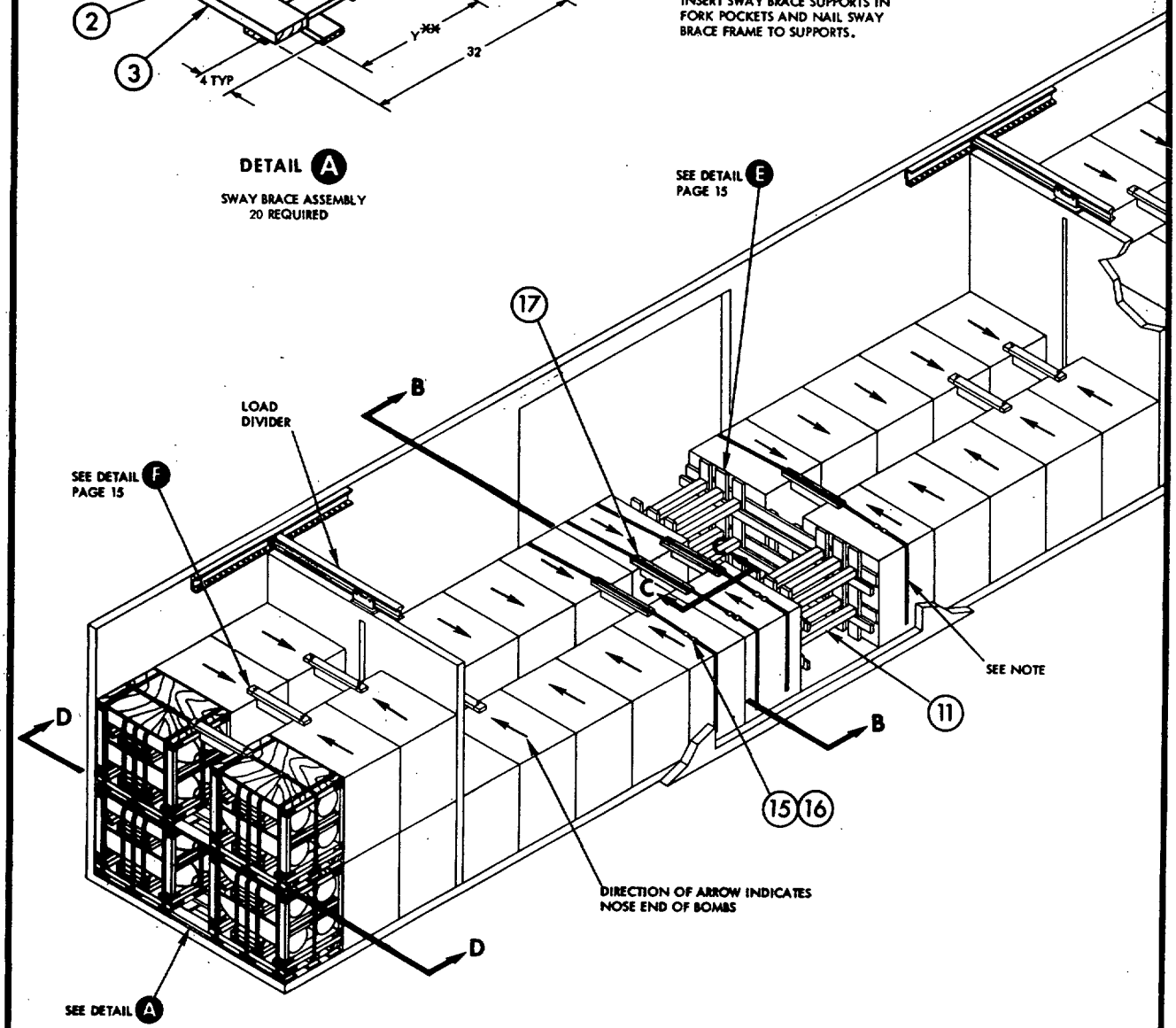
*W = DISTANCE BETWEEN PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

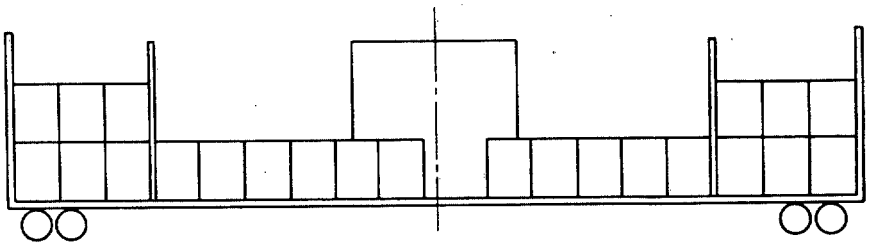
SWAY BRACE ASSEMBLY
20 REQUIRED



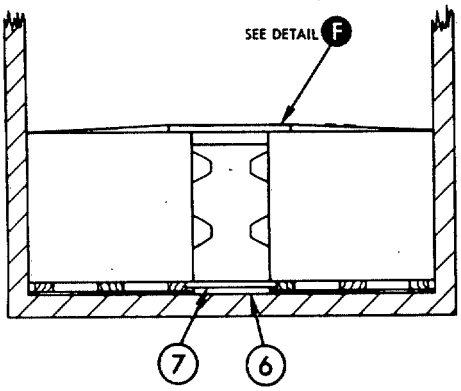
CARLOAD DATA

NUMBER OF UNIT LOADS	46
LOAD WEIGHT (APPROXIMATE)	143, 704 LBS
DUNNAGE WEIGHT (APPROXIMATE)	1, 179 LBS
CARLOAD WEIGHT (APPROXIMATE)	144, 883 LBS

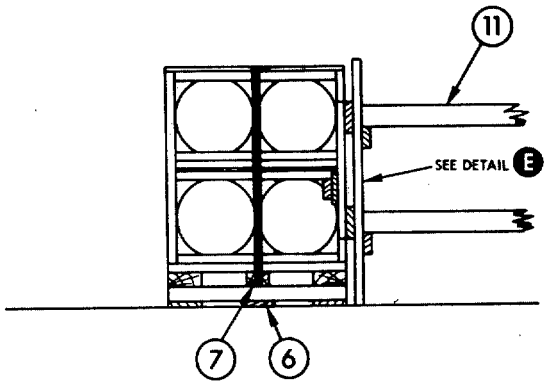
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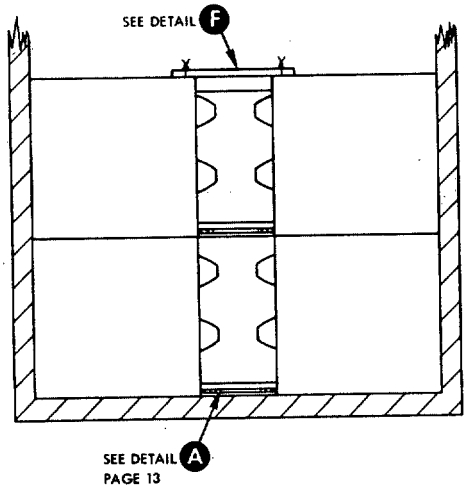
ELEVATION
LOADING PLAN



SECTION B-B

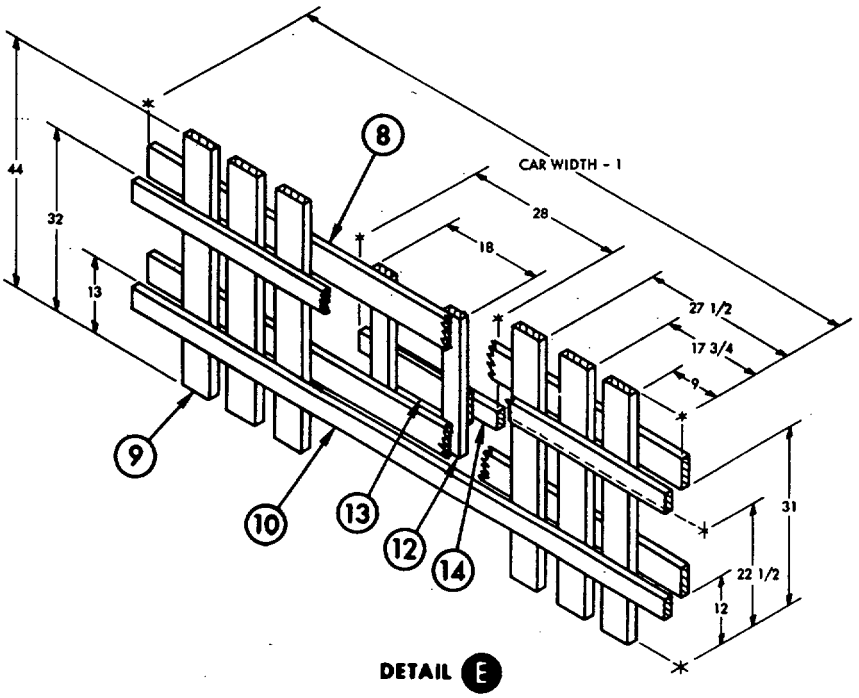


SECTION C-C

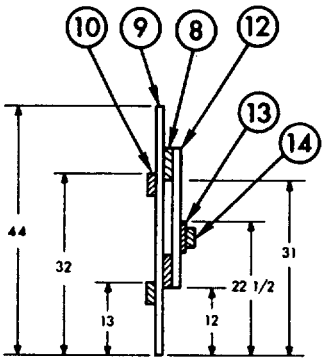


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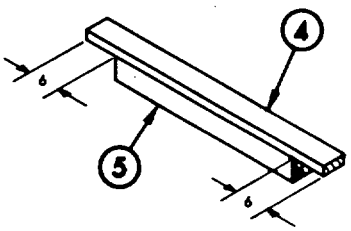
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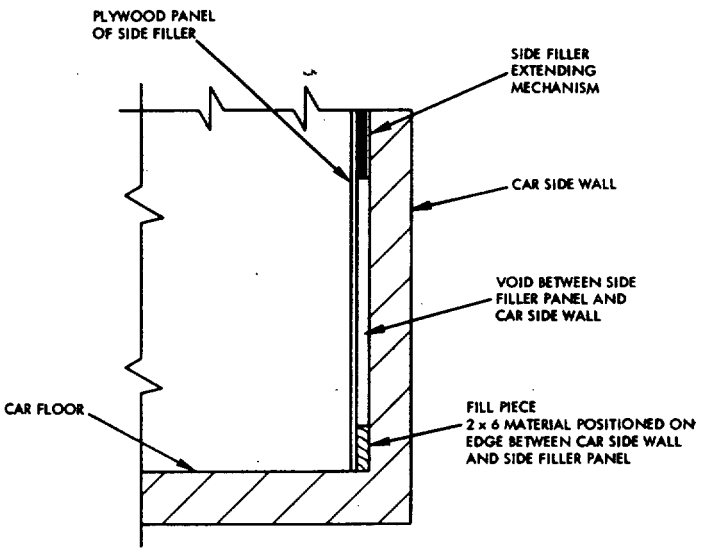
DETAIL E
CENTER GATE
2 REQUIRED



END VIEW



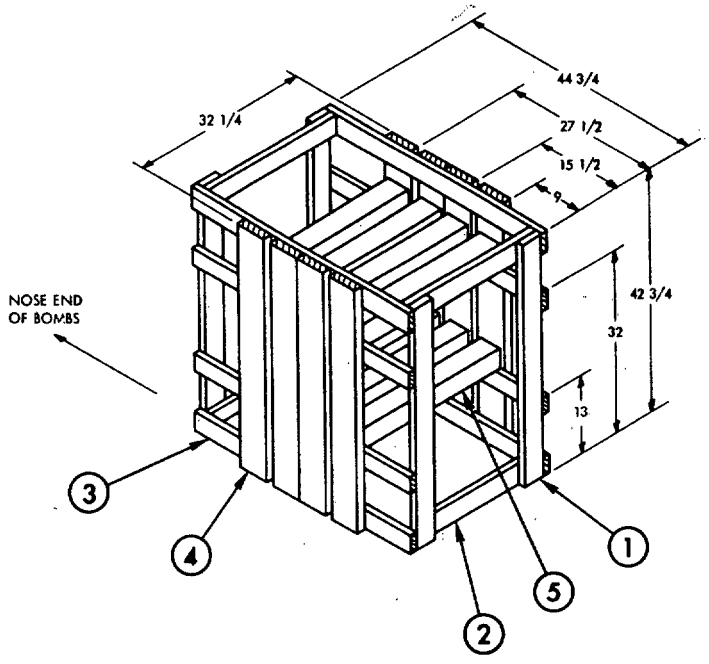
DETAIL F
TOP SWAY BRACE
13 REQUIRED



DETAIL G
SHOWING REINFORCEMENT FOR
ADJUSTABLE SIDE FILLERS

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DUMMY UNIT LOAD



NOTE:
TOENAIL STRUT TO VERTICAL WITH 16d NAILS, 3 EACH END (TOP AND BOTH SIDES), ALSO END NAIL THROUGH VERTICAL WITH 16d NAILS, 2 EACH END.

5	STRUT	4 x 4 x CUT TO FIT	8	4	SEE NOTE	
4	SIDE VERTICAL	2 x 6 x 42 3/4	8	3	3/JOINT	10d
3	LONGITUDINAL	2 x 4 x 44 3/4	8	1	2/JOINT	16d
2	CROSS BRACE	2 x 4 x CUT TO FIT	4	SEE 1	-	-
1	VERTICAL	2 x 4 x 42 3/4	4	2	3/JOINT	10d
PIECE NO.	DESCRIPTION	SIZE	NO.PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
LIST OF MATERIALS AND NAILING DATA						

REVIEW ACTIVITY
NAVY - SS, AS

PREPARING ACTIVITY:
NAVY - SS
(PROJECT NO. 8140-N243)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-STD-1325-171 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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